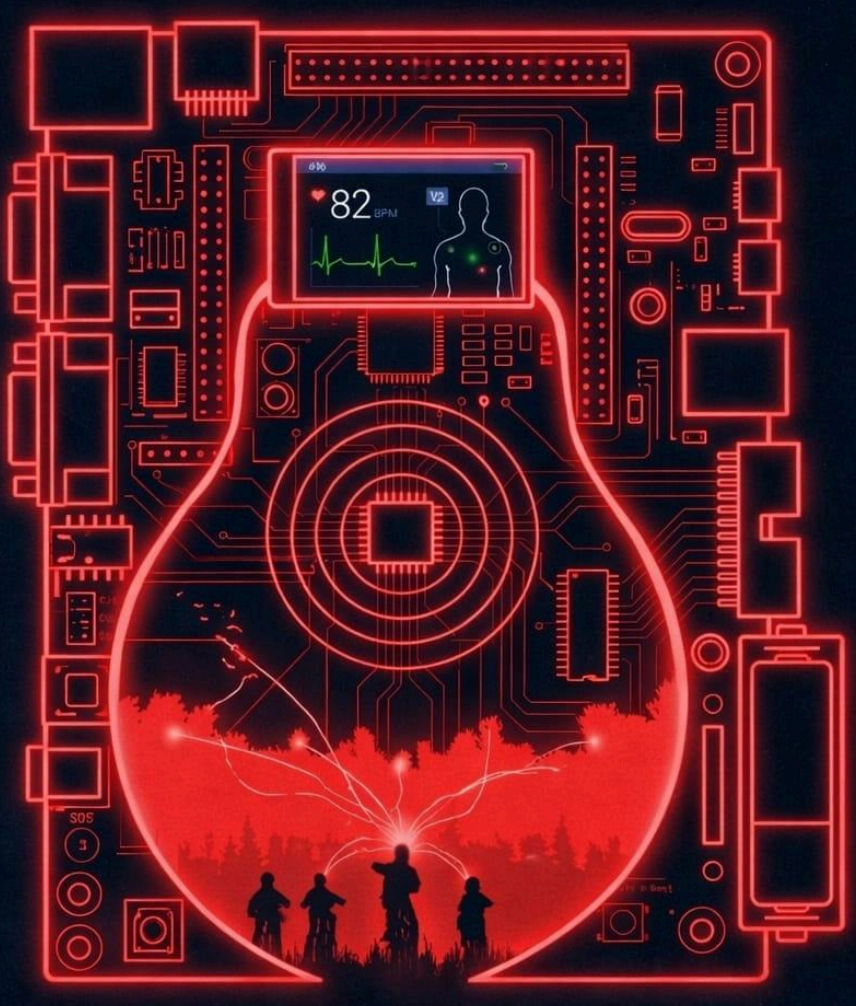
POWERED BY LOGIC GATES AND EXTRA-DIMENSIONAL FORCES

CIRCUIT PROTOCOL



RESONANCE 2026

Circuit Protocol

(PCB DESIGN)

OVERVIEW:

- Duration: 12 hours (9am to 7pm) (7pm to 9pm finishing and submissions)
- Team members: up to 3 members
- Software: KiCad (9.0 minimum)

PRIZE POOL : ₹10,000 (1st: ₹6,000 | 2nd: ₹4,000)
ENTRY : ₹450

RULES & REGULATIONS:

- Participants must use their own PCs to implement the problem statements on Kicad (version 9.0 minimum)
- A common name problem statement will be released a few minutes before the competition starts
- The participants must prepare a schematic design and a PCB layout of the given problem statement within the given time frame.
- Participants are allowed to
  a. Refer to datasheets of the manufacturers of the components being used.
  b. Use pre-made schematic symbols and footprints from open source providers in case it is absent from the inbuilt Kicad library.
- Participants are NOT allowed to
  a. Copy already existing designs (if any)
  b. Copy designs of other participants
  c. Use any AI.
  d. NO EXTERNAL PLUGINS ALLOWED. Eg Autorouting.
  e. Importing PCB files from a different software into KiCad

Doing so will result in a complete disqualification.
The co-ordinator's and judge's rulings are final and binding

BONUS PROBLEM STATEMENT:

Participants will be given three bonus problem statements each with more points to grab than the other. Those points will be proportional with the difficulty of the problem

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statement.

- Bonus PS 1: If completed, you will get +20 points
- Bonus PS 2: If completed, you will get +40 points
- Bonus PS 3: If completed, you will get +80 points

- It is NOT mandatory to do any of the bonus problem statements, but if opted for, then only ONE can be chosen.

- If any team attempts any of the BONUS problem statements AND fails to complete either the Main Problem Statement OR the Bonus problem statement OR both after opting for a bonus problem problem statement, points equivalent to the points of the chosen BONUS problem statement will be DEDUCTED from that team's final points tally.

Eg: Team X chooses Bonus problem statement 2 and fails to complete the main problem statement by the end, -40 points.

SUBMISSION GUIDELINE:

- Participants have to submit a report of their submission including their schematic, the PCB layout, detailing how the problem statement was implemented, along with how each section of the design was implemented including what all design practices were followed whilst implementing the same and also screenshots of the Electrical Rules Checker (ERC) window and the Design Rules Checker (DRC) window in PDF format.

- Participants have to submit a zip file of their KiCad project folder named in this format "TeamName_Zip" this folder must contain the manufacturing files like the drill files and the gerber files and also contain the aforementioned report in it, and the project folder must be named "TeamName" at the start of creation of the folder.

DESIGN CONSTRAINS:

- Participants must refer to the minimum design constraints from JLC PCB and use the relevant ones in their design https://jlcpcb.com/capabilities/pcb-capabilities.

NOTE: Lunch will be provided

CO-ORDINATORS:

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